



Effects of Irrigation Water Salinity and Organic Fertilizer Levels on Growth and Yield of Lettuce (*Lactuca sativa* L.)

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Abstract: A field experiment was conducted during the spring of 2024 to evaluate the effects of irrigation water salinity and organic fertilizer levels on the growth and yield of lettuce (*Lactuca sativa* L. ‘Parris Island Cos’). The experiment was arranged as a factorial within a split-plot design based on a randomized complete block design (RCBD) with three replications. Two irrigation water salinity levels were assigned to the main plots, namely fresh water (500 ppm) and saline water (4,500 ppm), while four levels of organic fertilizer (sheep manure), 0%, 1%, 2%, and 3%, were assigned to the subplots. The results showed that saline water irrigation generally produced higher numerical values for most growth and yield traits than fresh water under the conditions of this study. Among the tested fertilizer levels, the 1% organic fertilizer treatment gave the most favorable overall performance, particularly when combined with saline water. Under this treatment, plant height reached 19.1 cm, stem diameter 21.7 mm, leaf area 1.59 m², number of leaves per plant 75.8, plant weight 664.7 g, and total yield 53.2 t ha⁻¹. In conclusion, the findings suggest that a moderate rate of organic fertilizer may help maintain lettuce growth and productivity under saline irrigation conditions. These results also indicate that lettuce can be cultivated using saline water resources when supplemented with an appropriate level of organic fertilizer.

Keywords: Lettuce; Organic fertilizer; Sheep manure; Water salinity; Growth, Yield

1. Introduction

Lettuce (*Lactuca sativa* L.) is a significant vegetable crop belonging to the Asteraceae family and is highly valued for its nutritional and economic importance. It ranks among the top vegetable crops in many countries, including the United States, and is also a popular winter crop in Iraq and similar regions (Al-Chalabi et al., 2023; Han et al., 2019). Lettuce is rich in phytochemicals, vitamins A and C, antioxidants, minerals, and folic acid, which contribute to its importance as a major commercial crop (Kim et al., 2016).

The edible part of the lettuce plant is the leaf, which is commonly consumed fresh as a salad with other vegetables. According to food composition data, each 100 g of lettuce contains about 94.71% water, 3.69 g carbohydrates, 31 mg calcium, 23 mg phosphorus, 249 mg potassium, and 302 IU of vitamin A (USDA FoodData Central, 2024). In addition, lettuce contains small amounts of fat and several important mineral and bioactive compounds, including iron, magnesium, carotenoids, phenolic acids, and flavonoids. The nutritional value and medicinal benefits of lettuce leaves are mainly attributed to their characteristics as a natural and rich source of antioxidants, vitamins A, C, E, B9, and phytochemicals with potential health-promoting properties (Shi et al., 2022).

The increasing global concern over chemical residues in food production has driven consumer preference for organically grown produce. Organic agriculture relies on natural amendments, such as animal manure, to enhance soil fertility and crop quality while maintaining environmental sustainability. Concurrently, water scarcity and

salinity pose serious challenges to agriculture production making it necessary to explore the possibility of using saline water for irrigation, provided that its negative effects can be mitigated.

The salts in irrigation water and the soil solution have many direct and indirect effects on plant growth. Direct effects appear in the absorption of water by the plant, since increasing the salt concentration raises the osmotic pressure in the soil solution and reduces the plant's ability to absorb water. In addition, salts in the soil solution may lead to an imbalance in the absorption of nutrients needed by the plant, as the salts and their different ions may directly affect the plant through competition with some of the essential nutrients, sometimes reducing the absorption of important ions required for plant growth (El-Ramady et al., 2024). As for the indirect effects, they are mainly related to changes in physical and chemical traits of the soil, which subsequently affect plant growth and productivity (Lee et al., 2022).

Although salinity is generally recognized as a major factor limiting crop growth and productivity, several studies have shown that, under carefully controlled conditions, low to moderate salinity may not always be detrimental and can even stimulate growth or maintain productivity in some plant species (Li et al., 2021; Lou et al., 2025; Rouphael & Kyriacou, 2018). Under such conditions, salinity may act as a positive stimulus before reaching levels that inhibit physiological and developmental processes. Nevertheless, despite the considerable attention given to salinity over the past decades, the conditions under which it exerts beneficial rather than harmful effects remain insufficiently understood. In most crop species, including lettuce, salinity is still more commonly associated with osmotic stress, ion imbalance, reduced growth and yield, and physiological disorders such as tipburn related to oxidative stress (Carassay et al., 2012). Therefore, further research is needed to clarify the situations in which its negative effects may be minimized and whether moderate salinity can be utilized in a productive way under specific agronomic conditions.

Organic fertilization has received increasing attention as a sustainable agricultural practice for improving soil fertility, enhancing crop performance, and reducing dependence on chemical inputs (Bhatia & Sindhu, 2024; Kebalo et al., 2024). Previous studies have also shown that bio-organic fertilization can promote the growth and yield of vegetable crops (Aldolaimy et al., 2024; Masarirambi et al., 2012). Organic manures are known to improve soil physical and chemical properties, support microbial activity, and provide nutrients gradually over time, thereby creating a more favorable environment for plant growth. Masarirambi et al. (2012) showed that organic manures could improve vegetative growth and yield-related traits in lettuce, including leaf number, leaf area, and biomass accumulation. However, in lettuce, many previous studies have examined the effects of salinity stress (Di Mola et al., 2017; Ünükara et al., 2008; Yavuz et al., 2023) and organic fertilization (Alromian, 2020; Kebalo et al., 2024; Masarirambi et al., 2012), separately, whereas limited research has focused on the interaction between these two factors. More specifically, there is still insufficient information on whether a moderate rate of organic fertilizer can help maintain lettuce growth and productivity under relatively high irrigation water salinity, such as 4,500 ppm, and whether such fertilization can partially offset the adverse effects of saline irrigation.

Therefore, the present study aimed to assess the effects of irrigation water salinity and organic fertilizer rates on the vegetative growth and productivity of lettuce. The specific objectives were as follows:

1. To evaluate the potential of organic fertilizer (sheep manure) to promote growth, improve yield, and enhance the quality of lettuce under different irrigation water salinity conditions.
2. To investigate the effects of irrigation water salinity levels on the vegetative growth characteristics of lettuce plants.
3. To analyze the interaction between irrigation water salinity and organic fertilizer rates in their effects on vegetative growth and total yield of lettuce.

2. Materials and Methods

2.1 Study Site and Experimental Design

The experiment was carried out in the research fields of the Agricultural Technical College, Mosul, Iraq, during the spring season of 2024. The soil was prepared by plowing twice, after which decomposed sheep manure was incorporated according to the treatment specifications. A factorial experiment in a split-plot arrangement within a randomized complete block design (RCBD), with three replications, was used. The main-plots factor was irrigation water salinity, consisting of fresh water (500 ppm) and saline water (4,500 ppm). The sub-plots factor was the level of organic fertilizer, consisting of 0%, 1%, 2%, and 3% sheep manure. This resulted in eight treatment combinations and 24 experimental units. Each experimental unit consisted of three rows, with 50 cm between rows and 25 cm between plants within each row. Each unit contained 63 plants in total (21 plants per row). Seven plants from each row were marked as sampling plants. From these sampling plants, three representative plants from the center of each experimental unit were selected for the measurement of vegetative growth and yield traits.

2.2 Soil, Irrigation Water, and Climate Characteristics

Soil analysis was conducted before planting and after harvest, and the results were presented in Table 1. Soil

texture was determined before planting only. Climatic data recorded during the experimental period (April–June 2024) were presented in Table 2.

Table 3 showed the chemical analysis of sheep manure. The mineral composition of the sheep manure was determined in the central laboratory of the College of Agriculture and Forestry. The electrical conductivity of the irrigation water was 3.4 dS m⁻¹.

Table 1. Physical and chemical properties of the field soil

Parameter	Before Planting	After Harvest
Soil pH	6.5	6.2
Organic matter (%)	4.16	5.01
Electrical conductivity (dS m ⁻¹)	1.720	1.485
Nitrogen (ppm)	245	210
Phosphorus (ppm)	40.78	144.80
Potassium (ppm)	241.23	582.36
Particle-size distribution (%)		
Clay	20.8	—
Silt	36.7	—
Sand	42.5	—

Table 2. Monthly climatic data recorded during the experimental period (2024)

Month	Relative Humidity (%)	Average Temperature (°C)	Average Sunshine Duration (h day ⁻¹)	Rainfall (mm)
April	60	18.2	8	0
May	40	24.5	10	0
June	25	30.2	14	1

Source: Weather station in the experimental field.

Table 3. Chemical properties of sheep manure

Parameter	Value
Electrical conductivity (1:1 extract) (dS m ⁻¹)	5.94
Total nitrogen (g kg ⁻¹)	16
Total phosphorus (g kg ⁻¹)	1.8
Total potassium (g kg ⁻¹)	1.9
Organic carbon (g kg ⁻¹)	338.9
Organic matter (g kg ⁻¹)	584.2
C:N ratio	21:1

2.3 Plant Material and Crop Establishment

Seeds of the lettuce cultivar ‘Parris Island Cos’ were used in this experiment. This cultivar belongs to the long-head lettuce group (romaine lettuce) and is characterized by a cylindrical head shape, straight leaves, dark green color, smooth texture, prominent veins, and a compact head with a broad base. Head length usually ranges from 20 to 30 cm or more. The cultivar is easy to grow and is considered one of the standard lettuce cultivars in the United States.

Seeds were sown on 10 April 2024 according to the experimental design. Standard agronomic practices, including hoeing and weed control, were carried out as required. Harvesting took place between 5 and 10 June 2024, approximately 60 days after sowing.

2.4 Vegetative Growth and Yield Traits Measurement

Vegetative growth traits were measured at the end of the experiment, when the heads were harvested in the field. The following traits were recorded:

- (1) Plant height (cm): Plant height was measured from the point of contact with the soil to the growing tip.
- (2) Stem diameter (mm): The stem diameter was measured at the basal part of the plant after removing the leaves, using a Vernier caliper.
- (3) Leaf area (m²): Ten leaves of different sizes were selected from the plant. A known leaf area of 45 cm² was cut using a cork borer, and the fresh weight was recorded. Both the sampled area and the leaves were then dried in an oven at 65–72 °C for 72 h. After drying, the dry weights of the leaves and the known area were measured, and total leaf area was calculated using the ratio-and-proportion method.

(4) Number of leaves per plant: All leaves per head were counted except for the very small leaves located at the growing tip.

(5) Average total plant weight (g): Average total plant weight was determined at harvest by weighing representative whole plants from each experimental unit (Pastor-Arbulú & Rodríguez-Delfín, 2025).

(6) Total yield (t ha^{-1}): Total yield was calculated from the harvested weight of each experimental unit based on the area of the experimental unit and then converted to tons per hectare.

2.5 Statistical Analysis

The data were statistically analyzed according to the design used and the averages were compared using the Duncan's multiple range test at the 5% probability level (Al-Rawi & Khalaf Allah, 1980).

3. Results and Discussion

The interaction effects of irrigation water salinity and organic fertilizer on lettuce growth and yield were summarized in Table 4.

Table 4. Effect of irrigation water salinity and organic fertilizer levels on growth and yield parameters of lettuce

Water Salinity	Organic Fertilizer	Plant Height (cm)	Stem Diameter (mm)	Leaf Area (m^2)	Number of Leaves per Plant	Average Plant Weight (g)	Total Yield (t ha^{-1})
Saline water	Control	13.5 bc	22.5 a	0.86 a	64.4 abc	439.3 a	33.2 a
	1%	19.1 a	21.7 a	1.59 a	75.8 a	664.7 a	53.2 a
	2%	14.6 abc	26.4 a	1.01 a	64.8 abc	577.3 a	46.4 a
	3%	18.3 ab	22.6 a	0.99 a	59.2 bc	480.7 a	38.4 a
Main effect of saline water	—	16.4	23.3	1.11	66.1	540.5	42.8
Fresh water	Control	12.9 abc	22.1 a	0.69 a	52.8 ab	402.7 a	32.4 a
	1%	17.8 ab	25.3 a	1.23 a	77.5 a	613.8 a	51.2 a
	2%	14.4 c	21.6 a	1.09 a	68.9 c	545.1 a	43.6 a
	3%	15.3 abc	22.7 a	0.85 a	60.9 bc	414.7 a	33.2 a
Main effect of fresh water	—	15.1	22.9	0.97	65.0	494.1	40.1

Note: Means followed by the same letter within each column are not significantly different according to Duncan's multiple range test at the 5% probability level.

3.1 Plant Height

As shown in Table 4, the tallest plants (19.1 cm) were recorded under saline water irrigation combined with 1% organic fertilizer, and this value was statistically similar to that obtained with 1% organic fertilizer under fresh water irrigation (17.8 cm). In contrast, lower plant height values were observed in the control treatment and in some of the higher organic fertilizer levels, indicating that moderate organic fertilization was more effective than excessive application in promoting plant elongation. On average, saline water irrigation produced taller plants (16.4 cm) than fresh water irrigation (15.2 cm), suggesting that the salinity level used in this study did not suppress plant height. Rather, in combination with an appropriate level of organic fertilizer, it may have created conditions favorable for vegetative growth (El-Ramady et al., 2024). The superior response at the 1% organic fertilizer level may be attributed to improved nutrient availability and better soil physical conditions, which likely enhanced root activity and plant growth (El-Ramady et al., 2024). By contrast, increasing the organic fertilizer level to 2% or 3% did not further improve plant height, suggesting that the beneficial effect was greatest at the moderate level applied in this experiment. The results indicated that lettuce plant height responded positively to moderate organic fertilization, particularly when combined with saline water irrigation under the conditions of the current study.

3.2 Stem Diameter

The results shown in Table 4 indicated that stem diameter of lettuce was not significantly influenced by water salinity, organic fertilizer level, or their interaction, as all treatment means were followed by the same significance letter. But some numerical variation was also observed among treatments. Under saline water irrigation, stem diameter ranged from 21.7 to 26.4 mm, with the highest value recorded at the 2% organic fertilizer level, followed by the 3% treatment and the control. Under fresh water irrigation, values ranged from 21.6 to 25.3 mm, and the greatest stem diameter was obtained with 1% organic fertilizer. In terms of main effects, saline water produced a

slightly greater mean stem diameter (23.3 mm) than fresh water (22.9 mm), but this difference was not statistically significant. These findings suggested that stem diameter was relatively stable under the tested conditions and was less sensitive to treatment variation than plant height. The relatively close values among treatments may indicate that the salinity level applied in this study did not impose sufficient stress to adversely affect stem development. Likewise, the absence of a consistent increase with higher organic fertilizer levels suggests that stem diameter did not show a clear dose-dependent response to organic fertilization within the range tested.

3.3 Leaf Area

Data in Table 4 indicated that leaf area was not significantly influenced by water salinity, organic fertilizer level, or their interaction. However, numerical variation was observed among treatments. The highest leaf area was recorded under saline water irrigation combined with 1% organic fertilizer (1.59 m²), whereas the lowest value was found under fresh water without organic fertilizer (0.69 m²). In both irrigation conditions, the 1% organic fertilizer treatment produced the greatest leaf area, while higher fertilizer levels did not result in a further increase. The mean leaf area was also slightly higher under saline water (1.11 m²) than under fresh water (0.97 m²). These results suggest that moderate organic fertilization was more favorable for leaf expansion than higher application rates, although the differences were not statistically significant. They also indicate that the salinity level used in this study did not markedly limit leaf development.

The relatively higher leaf area recorded at the 1% organic fertilizer level may be associated with more favorable conditions for leaf expansion. Lettuce leaf area depends on both leaf number and the size of individual leaves, and these traits are known to be influenced by environmental factors such as light and temperature (AL-Bayati et al., 2019). Under the conditions of the present study, the improved nutrient availability provided by the moderate organic fertilizer level may therefore have supported better vegetative development.

3.4 Number of Leaves per Plant

The results of Table 4 showed that the highest values were recorded at the 1% organic fertilizer level under both saline and fresh water irrigation, reaching 75.8 and 77.5 leaves per plant, respectively, whereas relatively lower values were observed in the control and 3% organic fertilizer treatments. The mean number of leaves was also very similar under saline water (66.1) and fresh water (65.0), demonstrating that the salinity level used in this study did not significantly limit leaf production. Thus, it suggested that moderate organic fertilization was more favorable for leaf formation than either no fertilization or higher fertilizer application rates.

The relatively higher leaf number observed at the 1% organic fertilizer level may be related to more favorable conditions for vegetative growth. A moderate supply of organic fertilizer may have supported leaf initiation and expansion by providing nutrients gradually and by improving the growing environment. By contrast, the absence of a further increase at the higher fertilizer levels suggests that leaf production did not respond positively in a dose-dependent manner within the tested range. Therefore, 1% organic fertilizer appeared to be the most suitable level for promoting leaf development under the present experimental conditions. This suggested that a moderate organic fertilizer input might be sufficient to support leaf formation, whereas higher rates did not provide additional benefit. This trend is consistent with previous studies reporting that organic fertilization can improve vegetative growth in lettuce, although the response is not always proportional to increasing application rate (Alromian, 2020; Masarirambi et al., 2012).

3.5 Average Total Plant Weight

According to the results of treatments (Table 4), the value under saline water irrigation combined with 1% organic fertilizer was highest, reaching 664.7 g per plant, whereas the lowest value was observed in the unfertilized treatment under fresh water irrigation, at 402.7 g per plant. Under both irrigation conditions, the 1% organic fertilizer treatment produced higher plant weights than the control, while increasing the fertilizer level to 2% or 3% did not result in a further increase. In terms of the main effect of irrigation water, saline water produced a numerically greater mean plant weight (540.5 g) than fresh water (494.1 g). These results suggest that moderate organic fertilization was more favorable for biomass accumulation than either no fertilization or higher fertilizer application rates under the conditions of the present study.

The relatively greater plant weight recorded at the 1% organic fertilizer level may be associated with improved vegetative growth, as reflected in the higher number of leaves and larger leaf area observed in the same treatment combination. Since plant weight in lettuce is closely related to leaf development and overall canopy expansion, the increase in these growth traits may have contributed to the greater biomass accumulation. A moderate level of organic fertilizer may have provided a more balanced nutrient supply and improved the growing environment, thereby supporting better plant growth. By contrast, the absence of a further increase at the 2% and 3% fertilizer levels suggests that plant weight did not respond positively in a consistent dose-dependent manner within the tested

range.

These findings are generally consistent with previous studies showing that fertilization can enhance lettuce fresh weight compared with unfertilized treatments. For example, Zandvakili et al. (2019) reported that fertilization increased lettuce fresh weight relative to the unfertilized control. Similarly, Mokhtar et al. (2022) indicated that improvements in vegetative growth traits, such as leaf number, leaf area, and leaf fresh and dry weight, can contribute to greater plant biomass. In the present study, the higher plant weight observed at the moderate organic fertilizer level may therefore reflect more suitable growth conditions for lettuce plants.

3.6 Total Yield

The results presented in Table 4 indicated that total yield was not significantly affected by water salinity, organic fertilizer level, or their interaction, as all treatment means were followed by the same significance letter. However, numerical differences were observed among treatments. The highest total yield was recorded under saline water irrigation combined with 1% organic fertilizer, reaching 53.2 t ha⁻¹, whereas the lowest value was obtained in the unfertilized treatment under fresh water irrigation, at 32.4 t ha⁻¹, representing an increase of 64.2%. Under both irrigation conditions, the 1% organic fertilizer treatment produced the highest numerical yield, while the 2% and 3% fertilizer treatments did not result in a further increase. In terms of the main effect of irrigation water, saline water produced a slightly higher mean total yield (42.8 t ha⁻¹) than fresh water (40.1 t ha⁻¹), although this difference was not statistically significant.

These results suggest that moderate organic fertilization may be more favorable for yield formation than either no fertilization or higher fertilizer application rates under the conditions of the present study. The relatively greater yield observed at the 1% organic fertilizer level may be associated with improved vegetative growth, as reflected in the higher number of leaves, greater leaf area, and higher plant weight recorded in the same treatment. Since lettuce yield is closely related to vegetative development, the improvement in these growth traits may have contributed to the higher numerical yield values.

This interpretation is supported by previous studies showing that organic matter can improve soil fertility and quality by enhancing soil structure and increasing the availability of major and minor nutrients (Bhatia & Sindhu, 2024; Kebalo et al., 2024). In addition, humic substances derived from organic amendments may improve soil aggregate stability (Ai et al., 2023). Organic-based amendments and biologically active inputs may also enhance nutrient uptake and support plant growth under stress conditions (Shabaan et al., 2022). Therefore, the relatively higher yield observed in the fertilized treatments may reflect the beneficial role of organic amendments in supporting lettuce growth under both saline and fresh water irrigation. However, because the differences among treatments were not statistically significant, these findings should be interpreted as numerical trends rather than confirmed treatment effects.

4. Conclusions

The current study showed that lettuce could be grown under saline water irrigation (4,500 ppm) when combined with organic fertilizer. Among the tested fertilizer levels, 1% sheep waste generally produced the most favorable growth and yield performance. This treatment was associated with higher numerical values for plant height, leaf area, number of leaves per plant, plant weight, and total yield compared with the control and the higher fertilizer levels. These findings suggest that moderate organic fertilization may help maintain lettuce growth and productivity under saline irrigation conditions. In conclusion, the study indicates that combining saline water with organic amendments may be a promising strategy for lettuce production in areas with limited freshwater resources. Further research is needed to evaluate the long-term effects of this practice on soil properties and crop performance.

Author Contributions

Methodology, O.Y.H.; validation, A.A.; investigation, O.Y.H., M.S.A., O.A.F.A., and A.A.; fieldwork, M.A.B.; data organization, M.S.A.; writing—preparation of the original draft, A.A.; writing—review and editing, O.A.F.A. and A.M.S. All authors have read and approved the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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